# POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Numerical Methods

Semester: Fall

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

***Attempt all the questions.***

1. a) Solve $x^3+x^2-3x-3=0$ by secant method up to $8^{th}$ iteration. Assume that the error should be less than $10^{-4}$. 7

   b) Find the root of the equation $\log x - \cos x = 0$ correct to three decimal placed by using N-R method. 8

2. a) Define interpolation. From the following table, estimate the number of students who passed marks between 40 and 45: 8

| Marks | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 |
|---|---|---|---|---|---|
| No. of students | 30 | 40 | 50 | 38 | 31 |

   b) Fit cubic polynomial equations to the given data set and find the value of f (3.7) and f' (7.5). 7

| X | 2 | 4 | 7 | 9 |
|---|---|---|---|---|
| f(X) | 1 | 2 | 1 | 2 |

3. a) Integrate the following function by using Trapezoidal Rule, Simpson's $\frac{1}{3}$ rule and Simpson 3/8 rule. Take n = 6. $\int_0^{\frac{\pi}{2}} \sin x \, dx$ 8

   b) Integrate the given integral $\int_0^{\pi/2} \frac{\cos x}{\sqrt{1+\sin x}} dx$ 7

   Using Gauss quadrature Formula for n=2 and n=3

4. a) Find the inverse of the matrix, using Gauss Jordan method.. 8

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & -5 \\ -2 & -4 & -4 \end{bmatrix}$$

b) Find the largest Eigen-value and the corresponding Eigen-vector of the following square matrix using Power method.

$$\begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 4 \end{bmatrix}$$

5. a) Solve the following set of equations by using LU Crout method

$$3x+2y+z=10$$
$$2x+3y+2z=14$$
$$x+2y+3z=14$$

b) Apply R-K-4 method to solve y(0.2) for the given equation $\frac{d^2y}{dx^2}+x\frac{dy}{dx}-y$ given that y=1 and $\frac{dy}{dx}=0$ when x=0.

6. a) In a square bar with dimension of 3 inch × 3 inch, torsion function, $\phi$, can be obtained from the following P.D.E: $\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}=-2$ where $\phi=0$ on the outer boundary of the bar's cross-section. Subdivide the region into nine equal squares to form a mesh and find the values of $\phi$ in the interior nodes.

b) Consider second order initial value problem y"-4 y'+2y= $e^t$ sin (t) with y(0) = 0.4 and y'(0) = - 0.6, using Heun's find value of y(0.2) and y'(0.2).

7. Write short notes on: **(Any two)**

a) Taylor's series for solving ODE

b) Ill-Conditioned System

c) Classify the partial differential equation $U_{xx}+2U_{xy}+U_{yy}=0$

**POKHARA UNIVERSITY**

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*Attempt all the questions.*

1. a) What is the difference between the bisection method and false position even though both are bracketing methods? Find the real root of the given non-linear equation correct up to three decimal place using Newton Raphson method. 7

   b) Define error and write its different types in numerical methods with examples. If $x = 1.350253$ is rounded off to four significant digits, find the absolute and relative errors. 8

2. a) By using Least square method find the straight line that best fit the following data: 7

| | | | | | |
|---|---|---|---|---|---|
| x: | 1 | 2 | 3 | 4 | 5 |
| y: | 14 | 27 | 40 | 55 | 68 |

   b) Find the cubic spline interpolation formula for the following data: 8

| | | | | | |
|---|---|---|---|---|---|
| x | 1 | 2 | 3 | 4 | 5 |
| f(x) | 1 | 0 | 1 | 0 | 1 |

3. a) Evaluate $\int_4^{5.2} \log x \, dx$ from the following data 8

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| x | 4.0 | 4.2 | 4.4 | 4.6 | 4.8 | 5.0 | 5.2 |
| y | 1.3863 | 1.4351 | 1.4816 | 1.5261 | 1.5686 | 1.6094 | 1.6487 |

by using

i) Trapezoidal Method
ii) Simpson 1/3 Method
iii) Simpson 3/8 Method

b) Evaluate $\int_0^2 \frac{x^2+2x+1}{1+(x+1)^4}\,dx$ by using Gaussian Integration formula for n=2, n=3 and compare their values with exact solution.

4. a) Solve the following system of equations by using relaxation method correct to two decimal places.

$$9x-y+2z=9$$
$$x+10y-2z=15$$
$$2x-2y-13z=-17$$

b) Using Dollittle LU decomposition method, solve the following system equations:

$$3x+2y+z=10$$
$$2x+3y+2z=14$$
$$x+2y+3z=14$$

5. a) Use Runge-kutta of order four to find the solution of the given differential equation at x=1.5 taking a step size of h=0.25.

$$\frac{dy}{dx}+2y=x^2, y(1)=5$$

b) Find the solution of the given ordinary differential equation at x=0.5 using the step size of h=0.25 using Heun's method.

$$\frac{dy}{dx}+0.4y=3e^{-x}, y(0)=5$$

6. a) Determine the steady-state heat distribution in a thin square metal plate with dimensions 0.5 m by 0.5 m using n = m = 4. Two adjacent boundaries are held at 0°C, and the heat on the other boundaries increases linearly from 0°C at one corner to 100°C where the sides meet.

b) The following table gives the corresponding values of pressure and specific volume of superheated steam.

| V | 2 | 4 | 6 | 8 |
|---|---|---|---|---|
| P | 105 | 42.07 | 25.3 | 16.7 |

(i) Find the rate of change of pressure with respect to volume when V= 2.

(ii) Find the rate of change of volume with respect to pressure when P= 105.

7. Write short notes on: (Any two)

a) Ill-conditioned systems

b) Laplacian equation

c) Classification of Second Order Partial Differential Equation

**POKHARA UNIVERSITY**

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***Attempt all the questions.***

1. a) Find the root of the equation $f(x) = x^2-4x-10$ correct to three decimal places by using False Position method. 7

   b) Estimate the root of the equation $f(x) = xe^x - \cos x$ using Newton Raphson method correct to three decimal places. 8

2. a) From the following table estimate the number of student who obtained marks between 40 and 45. 7

| Marks | 30-40 | 40-50 | 50-60 | 60-70 |
|---|---|---|---|---|
| No of Students | 31 | 42 | 51 | 35 |

   b) From the following data given in the table below evaluate f (2.5) by using Lagrange method. 8

| x | 1 | 2 | 4 | 5 | 7 |
|---|---|---|---|---|---|
| f(x) | 1 | 1.414 | 1.732 | 2.00 | 2.6 |

3. a) Evaluate $\int_1^5 \frac{1}{x}dx$ by using Gaussian Integration formula for n=3 and compare the value with exact solution. 7

   b) Use the Romberg integration to find the solution correct upto three decimal places. 8

$$I = \int_0^1 \frac{1}{1+x^2}dx$$

4. a) Find the solution of the given simultaneous linear equation using Gauss Seidel method. 7

$6x_1-2x_2+x_3=11$

$-2x_1+7x_2+2x_3=5$

$x_1+2x_2-5x_3=-1$

b) Solve the following system of equations using Crout method.

$$x + y + z = 4, x + 4y + 3z = 8, x + 6y + 2z = 6$$

5. a) Using the Euler's(R-K I st order method) find an approximate value of y corresponding to x=1, given that dy/dx = X+Y and y= 1. When x=0, h = 0.1.

b) ApplyEuler's method to approximate value of $y(0.3)$ for the differential equation:

$$\frac{dy}{dx} = y + x, y(0) = 1.$$

6. a) Torsion on a square bar of size 15cm* 15cm. If two of the sides are held at $100^0$ C and the other two sides are held at $0^0$C. Calculate the steady state temperature at interior points. Assume a grid size of 5cm * 5cm.

b) Solve the Poisson equation $\nabla^2 f = 2x^2 + y$, over the square domain $1 \leq x \leq 4$, $1 \leq y \leq 4$, with f=0 on the boundary. Take step size in x and y, h=k=1.

7. Write short notes on: **(Any two)**

a) Ill-conditioned and Well-conditioned systems

b) Error in Numerical method

c) Cubic Spline.

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***Attempt all the questions.***

1. a) Explain in brief the errors in numerical calculations. 5
   b) Find a root of $3x+\sin x-e^x=0$ using 10
      i. One of the bracketing methods and
      ii. One of the non-bracketing methods

2. a) From the data given below, find the number of students whose weight is between 60 to 70. 8

| Weight in lbs | 0-40 | 40-60 | 60-80 | 80-100 | 100-120 |
|---|---|---|---|---|---|
| No. of students | 250 | 120 | 100 | 70 | 50 |

   b) Using the method of least squares, fit the curves $ax^2 + \frac{b}{x}$ to the following data. 7

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $y$ | -1.52 | 0.96 | 8.88 | 7.66 |

3. a) Use Romberg's method, to compute $\int_0^2 \frac{e^x+\sin x}{1+x^2}\,dx$ correct up to two decimal places. 8
   b) Estimate approximate derivative of $f(x) = x^2$ at x=1 for h=0.1, 0.2, 0.05, 0.01. Use first order difference method and find the respective error. 7

4. a) Apply the factorization method to solve the equation $3x+2y+7z=4$; $2x+3y+z=5$; $3x+4y+z=7$

   b) Using SOR method, solve the following system of

   $$4x+y+2z=4;\ 3x+5y+z=7;\ x+y+3z=3.$$

5. a) Find the largest eigen value and the corresponding eigen vector of the matrix

   $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ using power method

   b) Using the R-K Ist order method, find an approximate value of y corresponding to x=1, given that $\frac{dy}{dx}=\frac{y-x}{y+x}$ and $y=1$. When x=0, and h = 0.02. 7

6. a) Using the R-K method of fourth order, solve for y at x= 1.2, 1.4, from $\frac{dy}{dx}=\frac{2xy+e^x}{x^2+xe^x}$ given $x_0=1$, $y_0=0$. 8

   b) Solve the elliptic equation $U_{xx}+U_{yy}=0$ over a square mesh of side four units satisfying the following boundary conditions; $u(0,y)=0$ for $0\le y\le 4, u(4,y)=12+y$ for $0\le y\le 4; u(x,0)=3x$ for $0\le x\le 4, u(x,4)=x^2$ for $0\le x\le 4$ 7

7. Write short notes on: (Any two) 2×5

   a) Shooting Method

   b) Algorithm of Gauss Jordan method

   c) Algorithm of fixed point iteration method